

Worksheet: Transformers Fundamentals

Course: AI103

Student Name: _____

Part I: Multiple Choice Questions

1. What is the main difference between transformers and traditional RNNs?

- A. Transformers use self-attention mechanisms
- B. Transformers use convolutional layers
- C. Transformers use recurrent layers
- D. Transformers use fully connected layers

2. What is the purpose of the encoder in the transformer model?

- A. To generate an output sequence of symbols
- B. To map an input sequence of symbol representations to a sequence of continuous representations
- C. To perform attention mechanism
- D. To perform multi-head attention

3. What is the scaled dot-product attention mechanism used for in transformers?

- A. To counteract the effect of large dot products
- B. To increase the dimensionality of the input
- C. To decrease the dimensionality of the input
- D. To perform multi-head attention

4. How many heads are used in multi-head attention?

- A. 1
- B. 2
- C. h
- D. dk

5. What is the purpose of the softmax function in the attention mechanism?

- A. To increase the dimensionality of the input
- B. To decrease the dimensionality of the input
- C. To assign weights to the values
- D. To perform multi-head attention

6. What is the output of each sub-layer in the transformer model?

- A. $\text{LayerNorm}(x + \text{Sublayer}(x))$
- B. $\text{LayerNorm}(x - \text{Sublayer}(x))$
- C. $\text{LayerNorm}(x * \text{Sublayer}(x))$
- D. $\text{LayerNorm}(x / \text{Sublayer}(x))$

7. What is tokenization in the context of transformers?

- A. Converting text into numerical representations
- B. Converting numerical representations into text
- C. Performing attention mechanism
- D. Performing multi-head attention

8. What is the purpose of the decoder in the transformer model?

- A. To map an input sequence of symbol representations to a sequence of continuous representations
- B. To generate an output sequence of symbols one element at a time
- C. To perform attention mechanism
- D. To perform multi-head attention

9. What is the advantage of using transformers over traditional RNNs?

- A. Transformers are slower than RNNs
- B. Transformers are less accurate than RNNs
- C. Transformers can handle parallelization and reduce training time
- D. Transformers are more complex than RNNs

10. What is the mathematical representation of the attention function?

- A. $\text{Attention}(Q, K, V) = \text{softmax}(QK^T) * V$
- B. $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) * V$
- C. $\text{Attention}(Q, K, V) = \text{softmax}(QK^T + \sqrt{d_k}) * V$
- D. $\text{Attention}(Q, K, V) = \text{softmax}(QK^T - \sqrt{d_k}) * V$

11. What is the purpose of the compatibility function in the attention mechanism?

- A. To assign weights to the values
- B. To increase the dimensionality of the input
- C. To decrease the dimensionality of the input
- D. To perform multi-head attention

12. What is the advantage of using multi-head attention?

- A. It increases the computational cost
- B. It decreases the accuracy
- C. It allows the model to jointly attend to information from different representation subspaces
- D. It reduces the dimensionality of the input

13. What is the mathematical representation of multi-head attention?

- A. $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head1}, \dots, \text{headh}) * W^O$
- B. $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head1}, \dots, \text{headh}) / W^O$
- C. $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head1}, \dots, \text{headh}) + W^O$
- D. $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head1}, \dots, \text{headh}) - W^O$

14. What is the purpose of the projections in multi-head attention?

- A. To increase the dimensionality of the input
- B. To decrease the dimensionality of the input
- C. To perform attention mechanism
- D. To linearly project the queries, keys, and values

15. What is the advantage of using transformers for language modeling?

- A. Transformers are less accurate than RNNs
- B. Transformers are slower than RNNs
- C. Transformers can handle long-range dependencies
- D. Transformers are more complex than RNNs

16. What is the purpose of the softmax function in the scaled dot-product attention mechanism?

- A. To increase the dimensionality of the input
- B. To decrease the dimensionality of the input
- C. To assign weights to the values
- D. To perform multi-head attention

17. What is the mathematical representation of the dot product of the query with all keys?

- A. $q \cdot k = \sum_{i=1}^d q_i * k_i$
- B. $q \cdot k = \sum_{i=1}^d q_i + k_i$
- C. $q \cdot k = \sum_{i=1}^d q_i - k_i$
- D. $q \cdot k = \sum_{i=1}^d q_i / k_i$

18. What is the purpose of the scaling factor in the scaled dot-product attention mechanism?

- A. To increase the dimensionality of the input
- B. To decrease the dimensionality of the input
- C. To counteract the effect of large dot products
- D. To perform multi-head attention

19. What is the advantage of using transformers for text classification?

- A. Transformers are less accurate than RNNs
- B. Transformers are slower than RNNs
- C. Transformers can handle long-range dependencies
- D. Transformers are more complex than RNNs

20. What is the purpose of the decoder in the transformer model for machine translation?

- A. To map an input sequence of symbol representations to a sequence of continuous representations
- B. To generate an output sequence of symbols one element at a time
- C. To perform attention mechanism
- D. To perform multi-head attention

21. What is the advantage of using transformers for computer vision?

- A. Transformers are less accurate than CNNs
- B. Transformers are slower than CNNs
- C. Transformers can handle sequential data
- D. Transformers are more complex than CNNs

22. What is the purpose of the transformer model in natural language processing?

- A. To perform attention mechanism
- B. To perform multi-head attention
- C. To handle sequential data
- D. To handle non-sequential data

Part II: True / False

Write True or False for each statement.

1. _____ Transformers are a type of recurrent neural network (RNN).
2. _____ The transformer model uses self-attention mechanisms to draw global dependencies between input and output sequences.
3. _____ The scaled dot-product attention mechanism is used to increase the dimensionality of the input.
4. _____ Multi-head attention is used to allow the model to jointly attend to information from different representation subspaces.
5. _____ The transformer model is only useful for natural language processing (NLP) tasks.
6. _____ The attention function is computed as a weighted sum of the values.
7. _____ The softmax function is used to assign weights to the values in the attention mechanism.
8. _____ The transformer model uses convolutional layers to handle sequential data.
9. _____ The decoder in the transformer model generates an output sequence of symbols one element at a time.
10. _____ The transformer model is more complex than traditional RNNs.

Part III: Written Questions

1. What is the main difference between transformers and traditional RNNs?

2. How does the self-attention mechanism work in transformers?

3. What is the purpose of the scaled dot-product attention mechanism in transformers?

4. How does multi-head attention work in transformers?

5. What are some common applications of transformers?

6. How can transformers be used for text classification tasks?

7. What are some common misconceptions about transformers?

8. What is the mathematical representation of the attention function in transformers?

9. How does the transformer model handle sequential data?

10. What is the advantage of using transformers for natural language processing tasks?
